

Architectural Justification

Multi-Cloud Asset Cost Optimization Portal — Lab 3

Architecture Selection: We chose Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal.

Two Reasons

- Independent scaling: Billing CSV ingestion regularly processes multi-gigabyte AWS/Azure/GCP exports (NFR-001), which has very different resource needs than lightweight work like configuring a budget threshold or loading the cost dashboard. As separate services, the Billing Ingestion Service can be scaled or given dedicated compute independently of the other three, instead of the whole system scaling together.
- Fault isolation: If the Billing Ingestion or Asset Analysis service is slow or fails while parsing a large file, the Budget Alert service can still dispatch breach notifications and the Reporting & Dashboard service can still serve cached data. Keeping each Epic's workload (ingestion, analysis, alerting, reporting) in its own component prevents one slow path from taking down the whole portal.

Security Advantage

Splitting the system into services with their own interfaces to the Data Store (IBillingDataStore, IAssetDataStore, IBudgetDataStore, IReportDataStore) lets each service be granted least-privilege access to only the data it needs — for example, the Budget Alert service only needs threshold and spend totals, never raw billing line-items. This enforces NFR-002's role-based access control at the service boundary, rather than relying on one monolithic application where any code path could reach all billing data.

Performance Benefit

Because the Billing Ingestion Service is a separate, independently scalable component, it can be given the compute headroom needed to parse multi-gigabyte files and update the dashboard in under 2 minutes (NFR-001) without that heavy parsing workload slowing down the Reporting & Dashboard service's response time for a FinOps Analyst who just wants to view current spend.